INCH-POUND

MIL-DTL-83723/80C
20 October 2000
SUPERSEDING
MIL-C-83723/80B
27 December 1991

DETAIL SPECIFICATION SHEET

# CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), RECEPTACLE, (SOLDER FLANGE MOUNT, BAYONET COUPLING, SOLDER PIN CONTACT), (SERIES III, CLASSES H AND Y)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-83723.

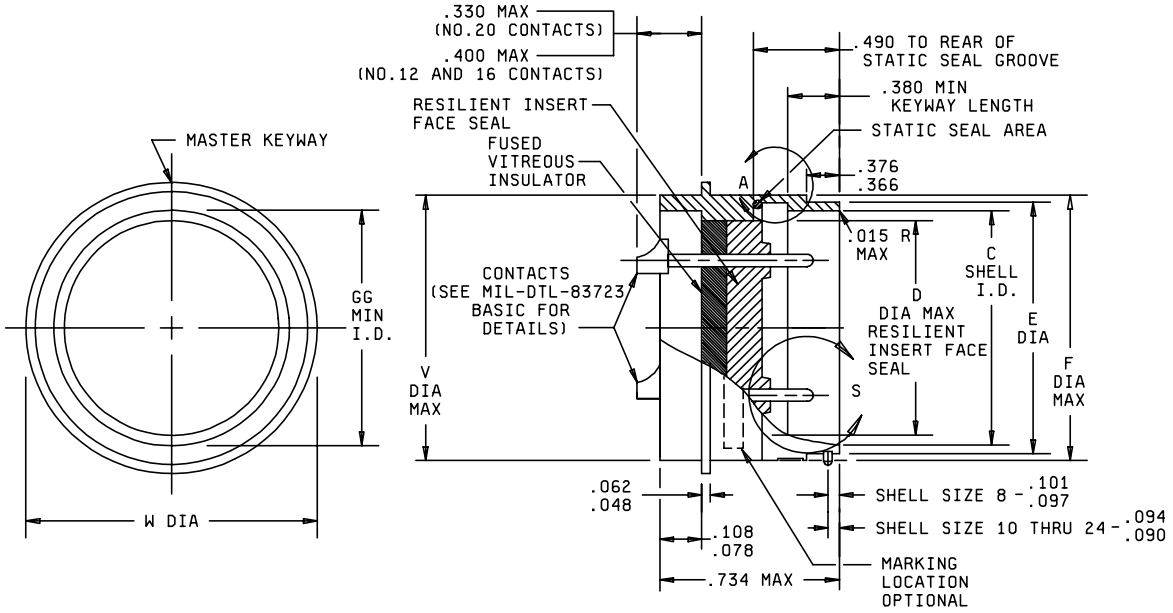



FIGURE 1. Dimensions and configuration.


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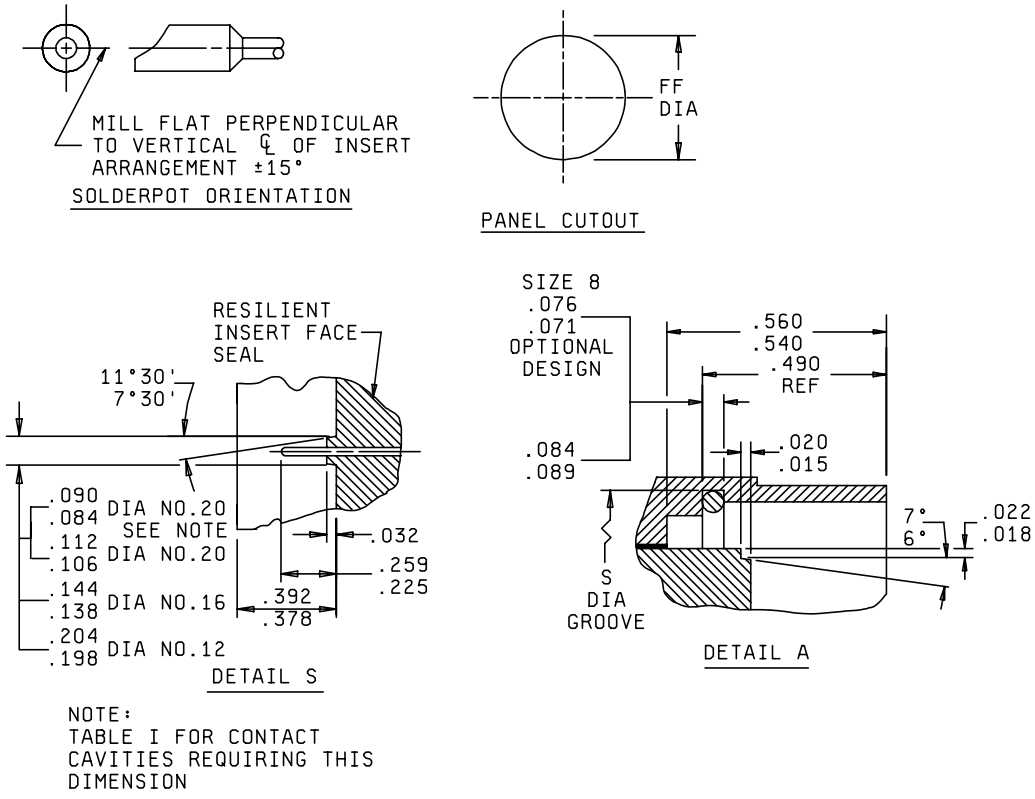



FIGURE 1. Dimensions and configuration - Continued.



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| Shell size | C | D | E | F | S +.005 -.000 | V | W | FF | GG |
|---|---|---|---|---|---|---|---|---|---|
| 8 | .428<br>.433 | .312 | .536<br>.531 | .561 | .508 | .500 | .760<br>.720 | .505<br>.515 | .406 |
| 10 | .530<br>.535 | .388 | .659<br>.654 | .696 | .636 | .562 | .860<br>.820 | .567<br>.577 | .443 |
| 12 | .700<br>.705 | .558 | .829<br>.824 | .875 | .806 | .750 | 1.065<br>1.025 | .755<br>.765 | .668 |
| 14 | .769<br>.774 | .627 | .898<br>.893 | .935 | .875 | .812 | 1.110<br>1.070 | .817<br>.827 | .668 |
| 16 | .896<br>.901 | .772 | 1.025<br>1.020 | 1.062 | 1.002 | .937 | 1.230<br>1.190 | .942<br>.952 | .763 |
| 18 | 1.002<br>1.007 | .860 | 1.131<br>1.126 | 1.187 | 1.108 | 1.062 | 1.360<br>1.320 | 1.067<br>1.077 | .862 |
| 20 | 1.127<br>1.132 | .985 | 1.256<br>1.251 | 1.312 | 1.233 | 1.187 | 1.450<br>1.410 | 1.192<br>1.202 | 1.108 |
| 22 | 1.252<br>1.257 | 1.110 | 1.381<br>1.376 | 1.437 | 1.358 | 1.312 | 1.610<br>1.570 | 1.317<br>1.327 | 1.204 |
| 24 | 1.377<br>1.382 | 1.235 | 1.506<br>1.501 | 1.562 | 1.483 | 1.437 | 1.730<br>1.690 | 1.442<br>1.452 | 1.388 |

FIGURE 1. Dimensions and configuration - Continued.



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| Inches | mm | Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|---|---|
| .003 | 0.08 | .330 | 8.38 | .636 | 16.15 | .901 | 22.89 | 1.251 | 31.78 |
| .004 | 0.10 | .366 | 9.30 | .654 | 16.61 | .935 | 23.75 | 1.252 | 31.80 |
| .005 | 0.13 | .376 | 9.55 | .659 | 16.74 | .937 | 23.80 | 1.256 | 31.90 |
| .015 | 0.38 | .378 | 9.60 | .668 | 16.97 | .942 | 23.93 | 1.257 | 31.93 |
| .018 | 0.48 | .380 | 9.65 | .696 | 17.68 | .952 | 24.18 | 1.312 | 33.32 |
| .020 | 0.51 | .388 | 9.86 | .700 | 17.78 | .985 | 25.02 | 1.317 | 33.45 |
| .022 | 0.56 | .392 | 9.96 | .705 | 17.92 | 1.002 | 25.45 | 1.320 | 33.53 |
| .032 | 0.81 | .400 | 10.16 | .720 | 18.29 | 1.007 | 25.58 | 1.327 | 33.71 |
| .048 | 1.22 | .406 | 10.31 | .734 | 18.64 | 1.020 | 25.91 | 1.332 | 33.83 |
| .062 | 1.57 | .428 | 10.87 | .750 | 19.05 | 1.025 | 26.04 | 1.358 | 34.49 |
| .071 | 1.80 | .433 | 11.00 | .755 | 19.18 | 1.062 | 26.97 | 1.360 | 34.54 |
| .076 | 1.93 | .443 | 11.25 | .760 | 19.30 | 1.065 | 27.05 | 1.376 | 34.95 |
| .078 | 1.98 | .490 | 12.45 | .763 | 19.38 | 1.067 | 27.10 | 1.377 | 34.98 |
| .084 | 2.13 | .500 | 12.70 | .765 | 19.43 | 1.070 | 27.18 | 1.381 | 35.08 |
| .089 | 2.26 | .505 | 12.83 | .769 | 19.53 | 1.077 | 27.36 | 1.382 | 35.10 |
| .090 | 2.29 | .508 | 12.90 | .772 | 19.61 | 1.108 | 28.14 | 1.388 | 35.26 |
| .094 | 2.39 | .515 | 13.08 | .774 | 19.66 | 1.110 | 28.19 | 1.410 | 35.81 |
| .097 | 2.46 | .530 | 13.46 | .806 | 20.47 | 1.126 | 28.60 | 1.437 | 36.50 |
| .101 | 2.57 | .531 | 13.49 | .812 | 20.62 | 1.127 | 28.63 | 1.442 | 36.63 |
| .106 | 2.69 | .535 | 13.51 | .820 | 20.83 | 1.131 | 28.73 | 1.450 | 36.83 |
| .108 | 2.74 | .536 | 13.61 | .824 | 20.93 | 1.132 | 28.75 | 1.452 | 36.88 |
| .112 | 2.84 | .540 | 13.72 | .827 | 21.01 | 1.187 | 30.15 | 1.483 | 37.67 |
| .138 | 3.51 | .558 | 14.17 | .829 | 21.06 | 1.190 | 30.23 | 1.501 | 38.13 |
| .144 | 3.66 | .560 | 14.22 | .860 | 21.84 | 1.192 | 30.28 | 1.506 | 38.25 |
| .198 | 5.03 | .561 | 14.25 | .862 | 21.98 | 1.202 | 30.53 | 1.562 | 39.67 |
| .204 | 5.18 | .562 | 14.27 | .875 | 22.23 | 1.204 | 30.58 | 1.570 | 39.89 |
| .225 | 5.72 | .567 | 14.40 | .893 | 22.68 | 1.230 | 31.24 | 1.610 | 40.89 |
| .259 | 6.58 | .577 | 14.66 | .896 | 22.76 | 1.233 | 31.32 | 1.690 | 42.93 |
| .312 | 7.92 | .627 | 15.93 | .898 | 22.81 | 1.235 | 31.37 | 1.730 | 43.94 |

NOTES:

1. Metric equivalents are given for general information only.
2. Dimensions are in inches.
3. Unless otherwise specified, tolerances are ± .005 (0.13 mm) for three place decimals and 01 (0.25 mm) for two place decimals and ±1 for all angles.
4. Dimensions apply after plating.
5. 63 all over in accordance with ANSI 46.1 unless otherwise specified.
6. Remove all burrs and sharp edges .003 to .005 R.
7. Diameters to be concentric within .004 TIR.
8. Diameter in the same cross-sectional plane to be concentric with each other within .003 TIR.

FIGURE 1. Dimensions and configuration - Continued.



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REQUIREMENTS:

Dimensions and configuration: See figure 1.

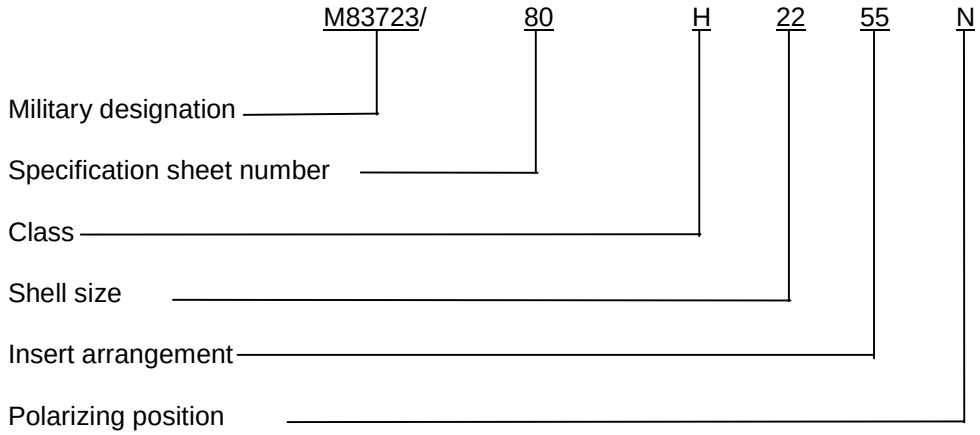
This connector mates with MIL-DTL-83723/75 and 77.

For insert arrangements, see MIL-STD-1554 and table I.

TABLE I. Insert arrangements requiring reduced diameters for raised seal barriers on contact cavities as indicated.

| Shell size | Insert arrangements | Contact cavities |
|---|---|---|
| 8 | 03, 98 | 1, 2, and 3 |
| 10 | 06 | 1, 2, 3, 4, 5, and 6 |

Part or Identifying Number (PIN) example:




CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI
Navy - EC
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5935-4281-11)



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